

1. Model Architecture

TinyHR is an end-to-end facial video model for estimating remote photoplethysmography (rPPG) signals and subsequently deriving heart rate from the predicted waveform. The architecture is organized as a lightweight convolutional pipeline that progressively transforms frame-difference information into spatial-temporal physiological representations. The model consists of four main components: (1) a Frame Difference Fusion Stem, (2) Spatial Patch Embedding, (3) a Multi-scale Temporal Feature (MTF) Block, and (4) a waveform Predictor Head. During inference, the predicted rPPG waveform is further processed using deterministic band-pass filtering and spectral analysis to obtain the final heart-rate estimate.

The input is a facial video clip containing 160 RGB frames at a spatial resolution of 128×128 . The model first emphasizes short-term temporal color variations between adjacent frames, then extracts compact spatial features for each frame. These features are aggregated spatially into a temporal sequence and processed by multi-scale temporal convolutional operations, allowing the network to capture physiological variations over different temporal contexts while retaining the original frame rate.

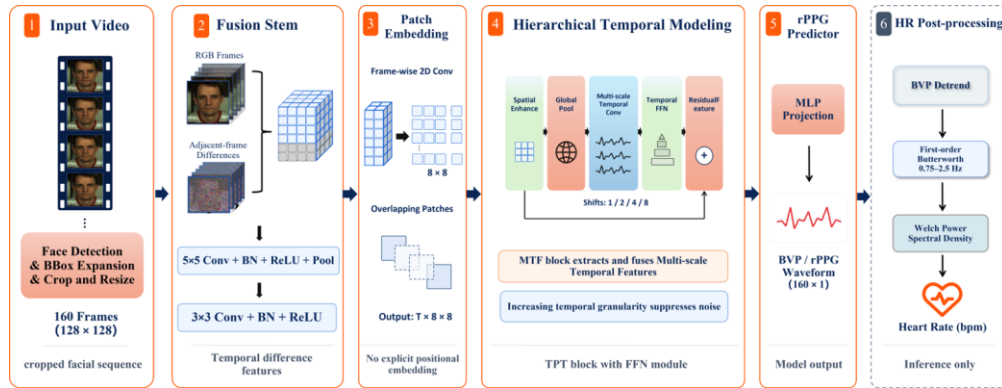


Figure 1. Model architecture of TinyHR.

1.2 Frame Difference Fusion Stem

The Frame Difference Fusion Stem focuses on the temporal color variations that carry rPPG information. For each frame, four adjacent-frame difference features are constructed using neighboring frames around the current time step. The four difference maps are concatenated along the channel dimension to form a 12-channel difference representation. This representation is processed exclusively by the difference branch, while the original RGB appearance branch is not used in the final model.

The 12-channel difference representation is first processed by a 2D convolution with a kernel size of 5 and stride 2, followed by batch normalization, ReLU activation, and max

pooling with stride 2. A subsequent 3×3 convolutional block further refines the feature representation. The resulting feature map contains 16 channels and has one quarter of the original spatial resolution. This design concentrates the early feature extraction process on frame-to-frame variations, which are more directly related to subtle pulsatile color changes in facial skin.

1.2 Spatial Patch Embedding

The output of the Difference Feature Stem is converted into a compact spatial representation on a frame-by-frame basis. A 2D convolution with a kernel size and stride of 4 reduces the spatial resolution from 32×32 to 8×8 while maintaining 32 feature channels. Batch normalization and ReLU activation are then applied. The temporal dimension is not downsampled during this operation; after processing individual frames, the features are rearranged into a video representation with dimensions corresponding to batch, channel, time, height, and width.

This frame-wise spatial embedding preserves the temporal sequence while reducing the spatial feature grid, providing a compact representation for subsequent spatiotemporal feature extraction.

1.3 Multi-Scale Temporal Feature Block

The core feature extraction module is a Multi-Scale Temporal Feature Block. Unlike an attention-based transformer block, this module models temporal dependencies using convolutional operations and explicit temporal sampling. The model uses a single such feature-extraction stage, without hierarchical temporal downsampling or multiple transformer stages. The block maintains the spatial feature map while deriving a compact temporal representation for physiological signal modeling.

First, a frame-wise spatial enhancement module processes each frame independently. The feature tensor is temporarily rearranged so that the batch and temporal dimensions are combined, and depthwise spatial convolution followed by pointwise convolution is applied to each frame. The enhanced frame features are then restored to the original spatiotemporal layout.

Next, spatial global average pooling aggregates the spatial dimensions of each frame, converting the feature map into a channel-wise temporal sequence. This sequence is processed by a multi-scale temporal convolution module with four parallel branches. The four branches use temporal shifts with different offsets, providing progressively larger temporal contexts. Each branch combines the shifted features with the current feature through channel concatenation and pointwise convolution. The outputs of the four branches are subsequently concatenated and fused by another pointwise convolution, producing a unified multi-scale temporal representation.

The resulting temporal features are further refined by a temporal feed-forward network composed of pointwise convolutional layers, batch normalization, and ReLU activation.

The refined temporal response is then broadcast back to the spatial feature map and added to the enhanced spatial representation. A residual connection from the block input is retained, followed by a ReLU activation. Through this process, the block jointly preserves frame-level spatial information and injects multi-scale temporal physiological cues into the feature representation.

1.4 Temporal Feature Feed-Forward Network

The temporal feed-forward network is used to increase the nonlinear representation capacity of the temporal features after multi-scale temporal aggregation. It expands the channel dimension, applies a sequence of pointwise convolution, batch normalization, and ReLU operations, and projects the representation back to the original channel dimension. A residual connection is used within the module so that the refined temporal representation retains information from the input sequence.

1.5 Waveform Predictor

After multi-scale temporal feature extraction, spatial global average pooling produces a compact temporal representation. The waveform predictor then applies two pointwise convolutional layers to map the temporal features to a single-channel signal. The predicted output is an rPPG waveform with one value for each video frame.

The neural network therefore predicts the rPPG waveform directly rather than the heart rate itself. Heart-rate estimation is performed after waveform prediction using the physiological signal-processing procedure.

2. Training Objectives

The network is optimized using complementary time-domain and frequency-domain objectives. The time-domain objective encourages the predicted waveform to reproduce the temporal morphology and phase of the reference BVP signal, while the frequency-domain objectives explicitly constrain the predicted signal toward the correct heart-rate distribution.

Let

$$\mathbf{S} = \{S_t\}_{t=1}^T$$

denote the reference BVP waveform and

$$\hat{\mathbf{S}} = \{\hat{S}_t\}_{t=1}^T$$

denote the predicted rPPG waveform for one training sample.

2.1 Time-Domain Loss

The time-domain objective is based on the negative Pearson correlation coefficient. First, the temporal means of the reference and predicted waveforms are defined as

$$\mu_S = \frac{1}{T} \sum_{t=1}^T S_t, \quad \mu_{\hat{S}} = \frac{1}{T} \sum_{t=1}^T \hat{S}_t.$$

The Pearson correlation coefficient is then formulated as

$$\rho(\mathbf{S}, \hat{\mathbf{S}}) = \frac{\sum_{t=1}^T (S_t - \mu_S) (\hat{S}_t - \mu_{\hat{S}})}{\sqrt{\sum_{t=1}^T (S_t - \mu_S)^2} \sqrt{\sum_{t=1}^T (\hat{S}_t - \mu_{\hat{S}})^2}}.$$

The time-domain loss is defined as

$$\mathcal{L}_{\text{time}} = -\rho(\mathbf{S}, \hat{\mathbf{S}}).$$

Because Pearson correlation is invariant to affine scaling, this objective focuses on the temporal morphology and phase consistency between the predicted and reference waveforms rather than their absolute amplitudes.

2.2 Frequency-Domain Cross-Entropy Loss

To explicitly constrain the physiological frequency of the predicted signal, the waveform is transformed into the frequency domain. Let f_s denote the video sampling rate and w_t denote a Hann window. The windowed discrete Fourier transform of the waveform is

$$\mathcal{F}(f) = \sum_{t=1}^T w_t \hat{S}_t e^{-j2\pi f t / f_s}.$$

The corresponding power spectral density (PSD) is

$$P_{\hat{S}}(f) = |\mathcal{F}(f)|^2.$$

Let $\mathcal{F}_{\text{HR}}$ denote the set of physiologically valid frequency bins. The normalized PSD is converted into a categorical probability distribution over heart-rate bins:

$$p_{\hat{S}}(f) = \frac{P_{\hat{S}}(f)}{\sum_{f' \in \mathcal{F}_{\text{HR}}} P_{\hat{S}}(f')}, \quad f \in \mathcal{F}_{\text{HR}}.$$

The reference BVP signal is processed in the same manner to obtain $P_S(f)$. The target heart-rate class is defined as the frequency bin corresponding to the dominant peak of the reference BVP spectrum:

$$f^* = \arg \max_{f \in \mathcal{F}_{\text{HR}}} P_S(f).$$

Let y_f denote the one-hot target distribution,

$$y_f = \begin{cases} 1, & f = f^*, \\ 0, & \text{otherwise.} \end{cases}$$

The frequency-domain cross-entropy loss is therefore

$$\mathcal{L}_{\text{CE}} = - \sum_{f \in \mathcal{F}_{\text{HR}}} y_f \log p_{\hat{s}}(f).$$

In the official implementation, the cross-entropy path uses heart-rate bins corresponding to 45–149 BPM. This objective complements the time-domain loss by explicitly encouraging the predicted waveform to concentrate its spectral energy around the physiological heart rate of the reference signal.

2.3 Heart-Rate Distribution Constraint

In addition to the categorical frequency-domain supervision, an HR-aware distribution constraint is introduced to measure the similarity between the spectral distributions of the reference and predicted signals.

Specifically, a Gaussian distribution is constructed around the dominant PSD frequency of each waveform. Let f_S^* and $f_{\hat{S}}^*$ denote the dominant frequency bins of the reference and predicted signals, respectively. Their corresponding Gaussian distributions can be written as

$$q_S(f) = \frac{1}{Z_S} \exp\left(-\frac{(f - f_S^*)^2}{2\sigma^2}\right),$$

and

$$q_{\hat{S}}(f) = \frac{1}{Z_{\hat{S}}} \exp\left(-\frac{(f - f_{\hat{S}}^*)^2}{2\sigma^2}\right),$$

where σ controls the distribution width and Z_S and $Z_{\hat{S}}$ are normalization constants.

The divergence between the reference and predicted HR distributions is measured using the Kullback-Leibler (KL) divergence:

$$\mathcal{L}_{\text{KL}} = D_{\text{KL}}(q_S \parallel q_{\hat{S}}) = \sum_{f \in \mathcal{F}_{\text{HR}}} q_S(f) \log \frac{q_S(f)}{q_{\hat{S}}(f)}.$$

This constraint provides an HR-oriented measure of the agreement between the predicted and reference spectral distributions.

2.4 Overall Training Objective

The complete training objective combines the time-domain waveform loss and the frequency-domain supervision:

$$\mathcal{L} = \lambda_{\text{time}} \mathcal{L}_{\text{time}} + \lambda_{\text{CE}} \mathcal{L}_{\text{CE}} + \lambda_{\text{KL}} \mathcal{L}_{\text{KL}},$$

where λ_{time} , λ_{CE} , and λ_{KL} are weighting coefficients set as 0.2, 1.0, 1.0, respectively.

The three terms provide complementary supervision. Specifically, $\mathcal{L}_{\text{time}}$ encourages *temporal waveform consistency*, $\mathcal{L}_{\text{CE}}$ explicitly identifies the correct heart-rate frequency bin, and $\mathcal{L}_{\text{KL}}$ encourages agreement between the predicted and reference HR distributions. Together, they guide the network to generate an rPPG waveform that is simultaneously temporally consistent and physiologically meaningful.

3. Heart-Rate Estimation During Inference

The output of the neural network is an rPPG waveform rather than a heart-rate value. Therefore, a deterministic physiological signal-processing pipeline is applied after waveform prediction.

Given the predicted waveform $\hat{S}$, a second-order Butterworth band-pass filter with cutoff frequencies of 0.75 Hz and 2.5 Hz is first applied to suppress frequencies outside the expected physiological heart-rate range. The power spectral density is then estimated using Welch's method. The dominant frequency is obtained as

$$f^* = \arg \max_f P(f),$$

where $P(f)$ denotes the estimated PSD of the filtered rPPG signal. Finally, the heart rate is calculated as

$$\text{HR} = 60f^*,$$

where HR is expressed in beats per minute (BPM) and f^* is measured in Hz.

4. Training Configuration and Experimental Results

The model was trained using a combination of multiple rPPG datasets to improve the diversity of subjects, recording conditions, and physiological signals. The training set consisted of four datasets: **VIPL-HR V1**, **VIPL-HR V2**, **V4V**, and **MCD-rPPG**.

Specifically, VIPL-HR V1 contains 85 subjects and 1,883 videos, while VIPL-HR V2 contains 500 subjects and 2,498 videos. The V4V dataset contains 103 subjects and 726 videos. For MCD-rPPG, only front-facing videos were retained for training, resulting in 600 subjects and 1,200 videos.

The resulting training set therefore covers a relatively broad range of subjects and recording conditions, providing diverse training samples for learning robust rPPG representations.

Dataset	Subjects	Videos	Usage
VIPL-HR V1	85	1,883	Training
VIPL-HR V2	500	2,498	Training
V4V	103	726	Training
MCD-rPPG	600	1,200	Training; front-facing videos only

The model was trained with a batch size of 4. The initial learning rate was set to 0.005, and the **OneCycleLR** learning-rate scheduling strategy was adopted to dynamically adjust the learning rate throughout training.

The main training configuration is summarized as follows:

Configuration	Setting
Batch size	4
Initial learning rate	0.005
Learning-rate scheduler	OneCycleLR
Training datasets	VIPL-HR V1, VIPL-HR V2, V4V, MCD-rPPG

To evaluate the generalization capability of the trained model, experiments were conducted on the **VIPL-HR V1 test set**, which contains 22 subjects and 485 videos. Importantly, the test set was used only for evaluation and was not included in the training data. The model achieved a Mean Absolute Error (MAE) of 3.88 BPM on the VIPL-HR V1 test set.